

Winter project

MOON EVOLUTION ODYSSEY

Analysing habitability potential on icy moons
Europa and Enceladus by studying their
geophysical characteristics and tidal heating

Mentors:

Chaitanya

Chhavi

Devansh

Pearl

Tarushi

Winter project

MENTEES

Anubha Jaiswal

Atharv Dubey

Priyam

Vansh Sejpal

Srushti Jagtap

Aditi Yadav

Krishna Soni

Manasvi Gangwar



Classification of moons:

- **on the basis of physical properties:**
Regular, Irregular, shepherd and trojan
- **on the basis of surface composition:**
Rocky, Icy and Mixed-composition
- **on the basis of Geological activity:**
Geologically Active and Inactive
- **on the basis of atmosphere:**
With and without atmosphere
- **on the basis of origin:**
Primordial and Captured



Mercury and Venus

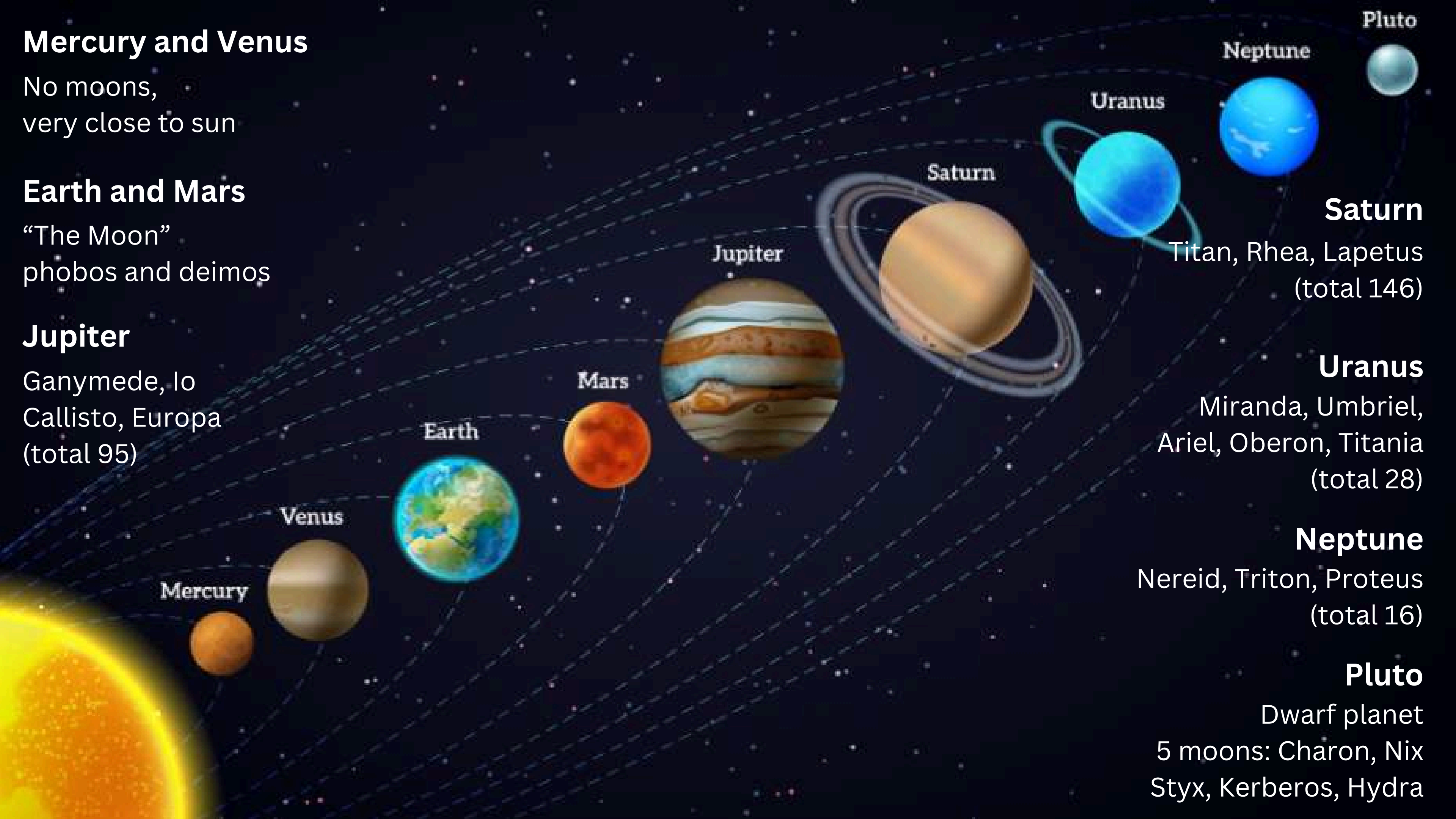
No moons,
very close to sun

Earth and Mars

“The Moon”
phobos and deimos

Jupiter

Ganymede, Io
Callisto, Europa
(total 95)



Saturn

Titan, Rhea, Lapetus
(total 146)

Uranus

Miranda, Umbriel,
Ariel, Oberon, Titania
(total 28)

Neptune

Nereid, Triton, Proteus
(total 16)

Pluto

Dwarf planet
5 moons: Charon, Nix
Styx, Kerberos, Hydra

Theories on origin of moon:

Fission Theory:

the Moon was originally part of the Earth but was torn free

Co-creation theory:

The two closely spaced proto-planets formed with the smaller one in orbit. They grew and eventually became what we know as Moon and Earth

Capture theory:

the Moon was formed somewhere else in the solar system and was later captured by the gravitational field of the Earth

Impact theory:

Earth was hit by a Mars sized object. Most of the material reformed as the Earth, and some material orbited the Earth and eventually formed the Moon



FEATURES OF MOON

Craters

Craters form from meteoroid, asteroid, or comet impacts, varying in size. Notable craters include Tycho, Copernicus, and Clavius. Many craters have central peaks from surface rebound post-impact.

Rays

The rays surrounding the craters are material ejected during the impacts that formed the craters. Only the rays of the more “recently” formed craters maintain their features. Older rays have been degraded because of meteor impacts.

Maria

The maria on the Moon are vast, dark plains formed by volcanic eruptions, less cratered than highlands, covering 17% of the Moon's surface. They consist mainly of basalt and include notable examples like Mare Imbrium, Mare Serenitatis, and Mare Tranquillitatis.

Lunar Highlands

The lunar highlands cover 83% of the moon's surface, characterized by rugged, mountainous terrain and composed mainly of anorthosite, an igneous rock abundant in plagioclase feldspar.

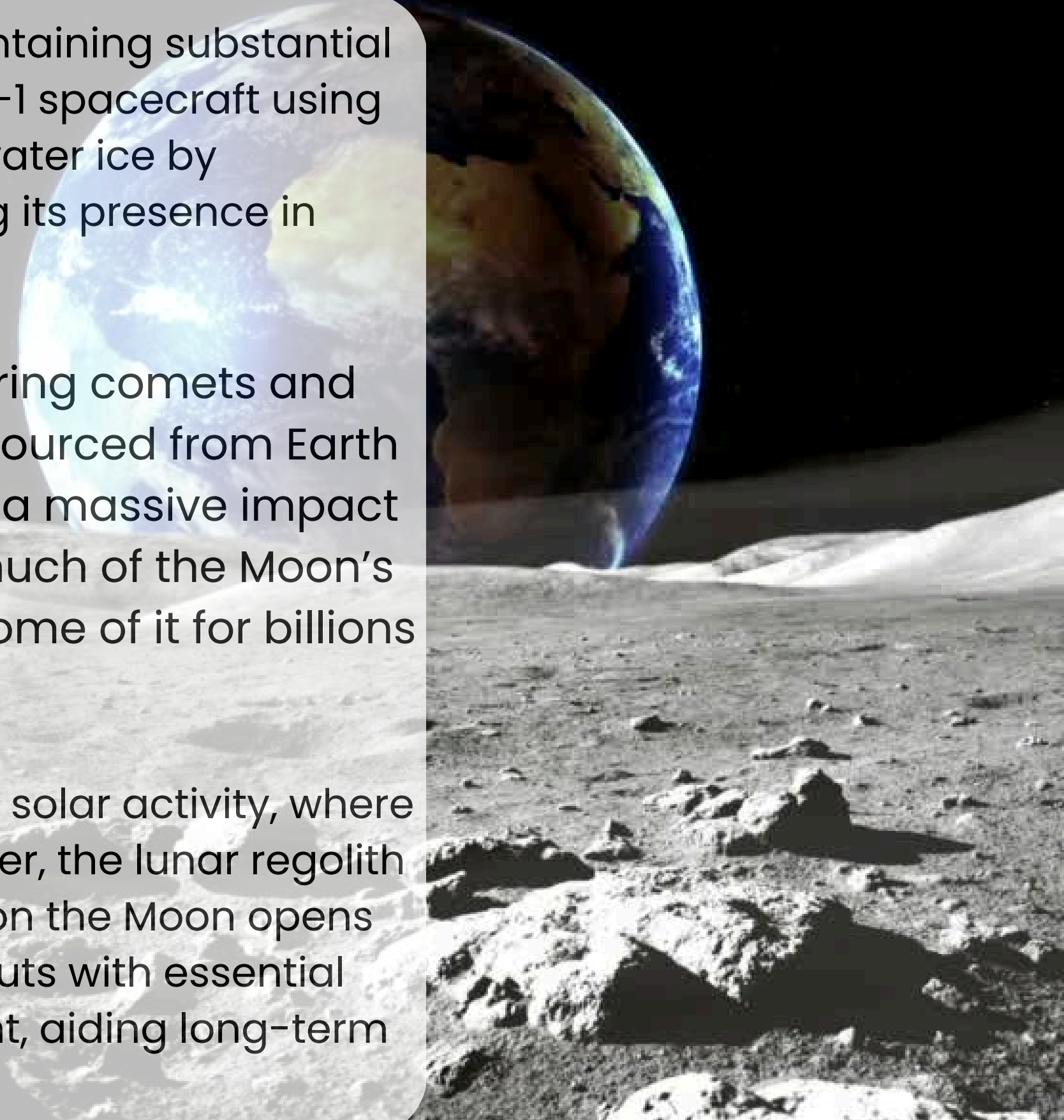
Lunar Cataclysm

The lunar cataclysm, also known as the Late Heavy Bombardment, is a hypothesized astronomical event that occurred approximately 3.8 to 4.1 billion years ago. It's thought to have been a period of intense asteroid and cometary collisions that severely heated the Moon's crust and metamorphosed its rocks.



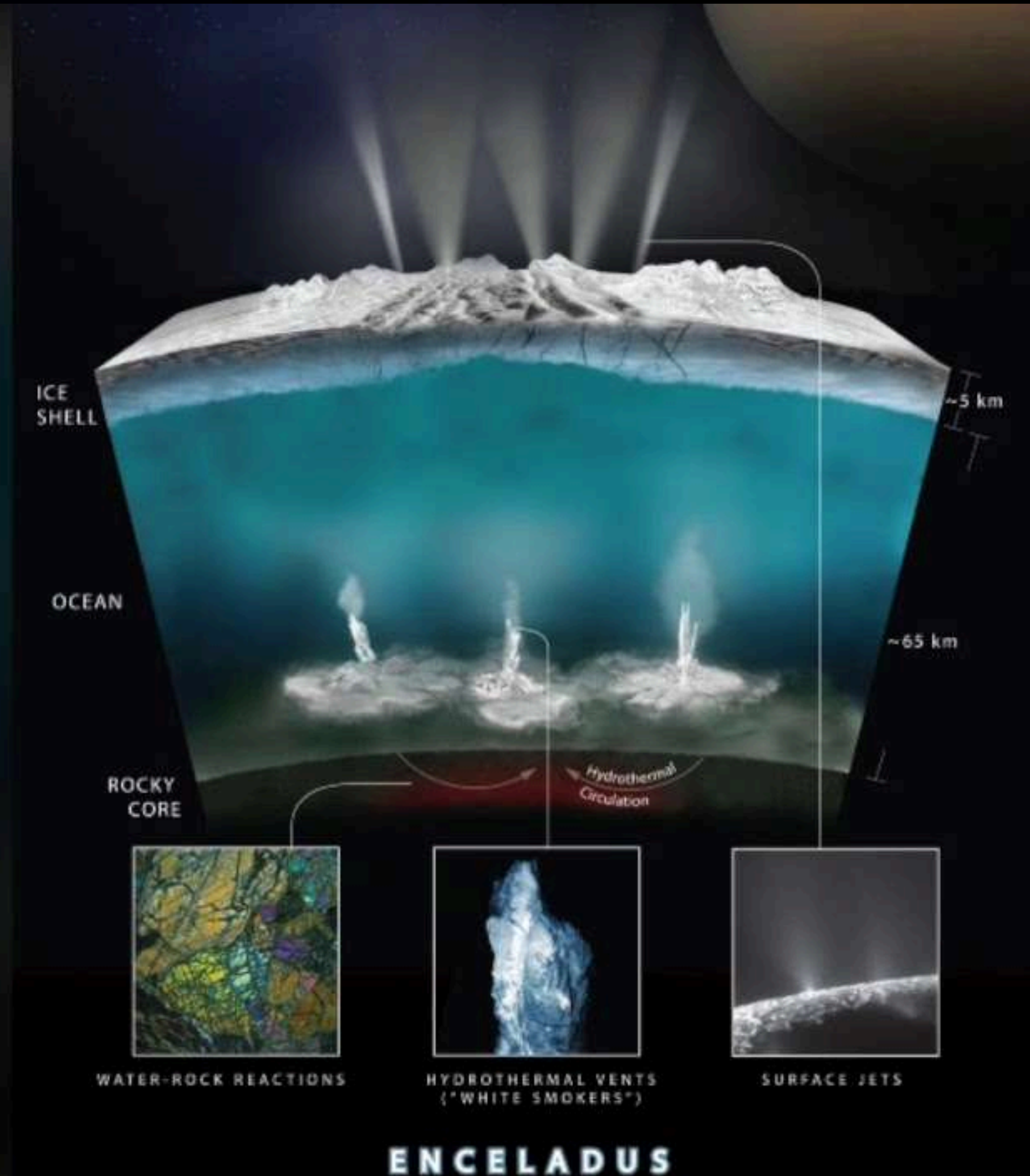
Water Ice on the Moon: Discovery, Origins, and Future Exploration

- The Moon's poles host permanently shadowed regions (PSRs) containing substantial amounts of water ice, a discovery made by India's Chandrayaan-1 spacecraft using the Moon Mineralogical Mapper (M3). This instrument identified water ice by analyzing how the lunar surface absorbs infrared light, confirming its presence in polar craters.
- The origins of this water are linked to impacts by water-bearing comets and asteroids over the Moon's history, as well as possibly being sourced from Earth during the Moon's formation, which is thought to result from a massive impact between the early Earth and another celestial body. While much of the Moon's water has evaporated over time, the PSRs have preserved some of it for billions of years due to their cold, shadowed conditions.
- Outside the polar regions, small amounts of water are created by solar activity, where protons from the Sun interact with oxygen in the lunar soil. However, the lunar regolith remains far drier than Earth's deserts. The discovery of water ice on the Moon opens possibilities for future space missions, as it could provide astronauts with essential resources like drinking water, breathable air, and rocket propellant, aiding long-term space exploration efforts.



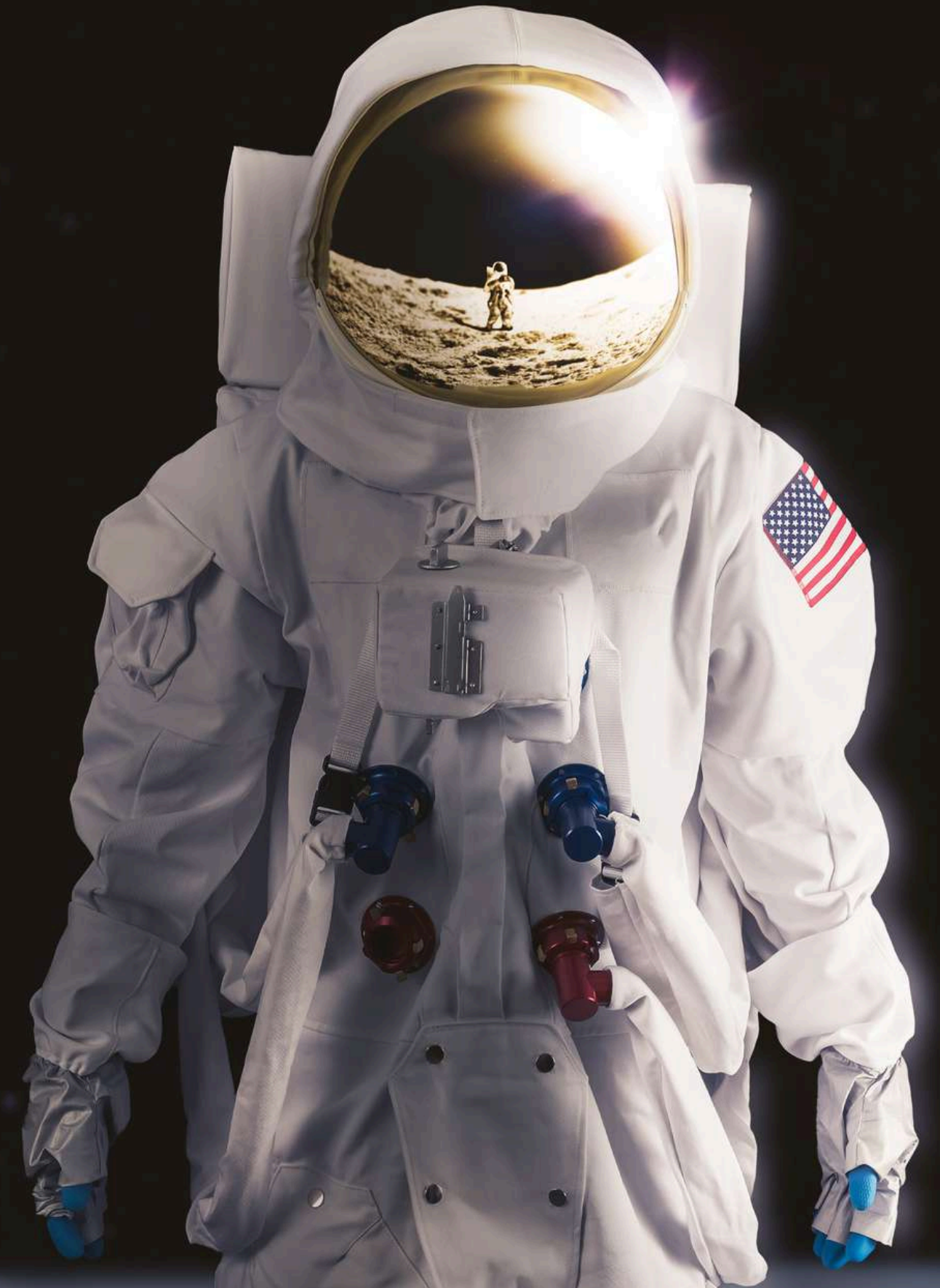
Enceladus: A Promising Candidate for Extraterrestrial Life

- Enceladus, Saturn's sixth-largest moon, is a key target in the search for extraterrestrial life due to its reflective surface, active geysers, and subsurface ocean.
- Discovered in 1789, it shows geological renewal and ejects water vapor, organic molecules, and other compounds into space, sustaining Saturn's E-ring.
- Cassini spacecraft data confirmed the ocean's potential habitability, with hydrothermal activity akin to Earth's deep-sea vents. Future missions aim to explore its geology and potential for life, advancing planetary science and the search for life beyond Earth.

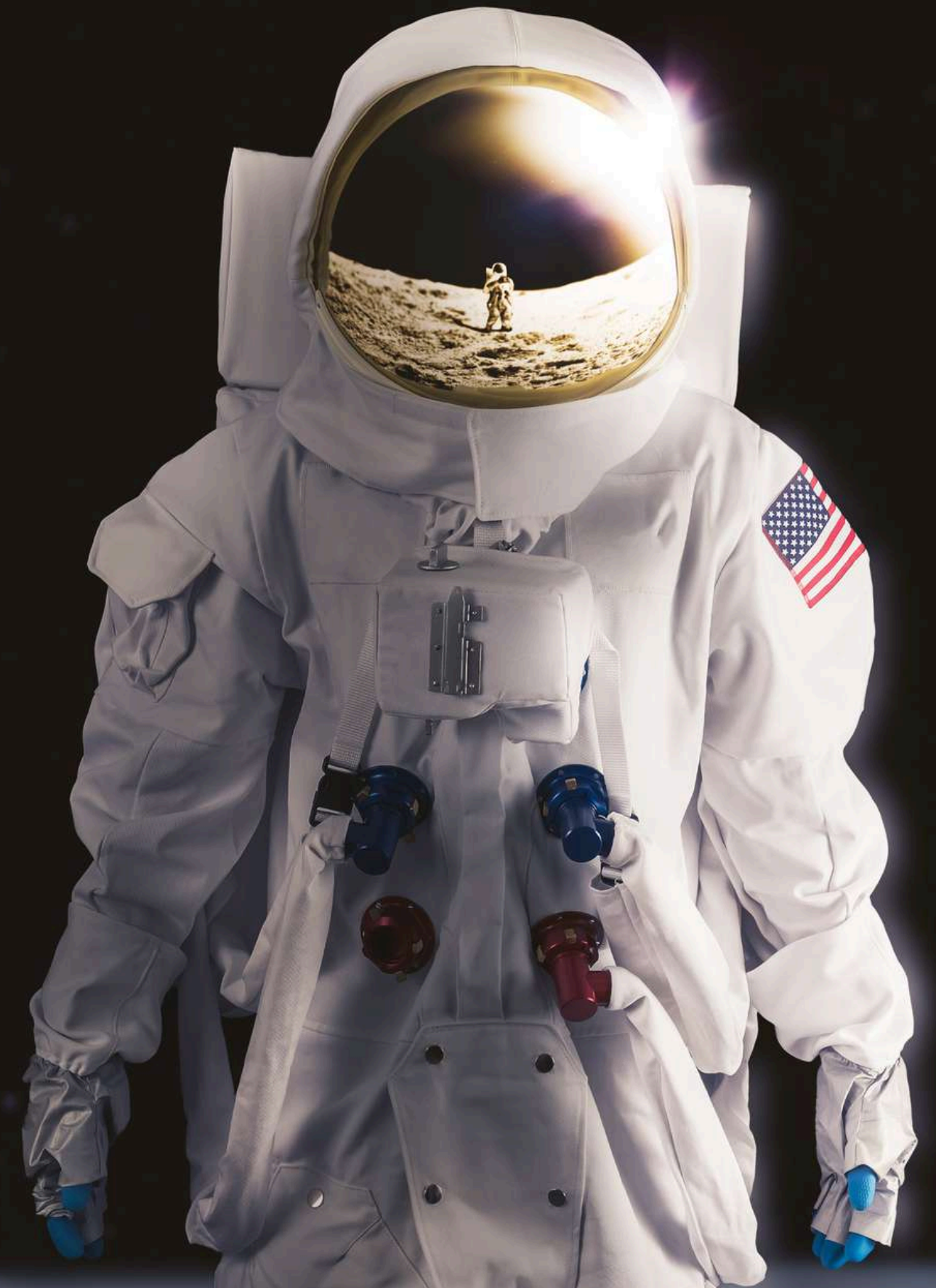


Exploring Europa: Potential for Life and Upcoming Missions

- Europa, one of Jupiter's moons, is a focus of scientific interest due to its potential to harbor life.
- Data from the Galileo spacecraft indicate Europa has a structure similar to Earth's, with a silicate rock composition, an iron core, and a rocky mantle.
- A subsurface ocean beneath its icy crust is hypothesized based on magnetic field fluctuations, raising the possibility of extraterrestrial life.
- The moon's surface is covered with cracks, believed to result from tidal forces caused by Europa's orbit around Jupiter. These forces may cause the subsurface ocean to rise and fall, creating surface fractures.
- Sporadic plumes, likely venting material from the ocean, suggest connections to the mantle, similar to Earth's hydrothermal vents, which can sustain life.



- A 2016 study revealed that Europa produces 10 times more oxygen than hydrogen, indicating chemical reactions could provide energy for life without solely relying on tidal heating.
- NASA's Europa Clipper mission will explore the moon's potential habitability by studying its surface, subsurface ocean, and chemical makeup.
- Equipped with advanced instruments, the mission aims to assess whether Europa's conditions could support life and deepen our understanding of extraterrestrial environments.
- Additionally, researchers have developed a Monte Carlo exosphere model to analyze how Europa's water plumes affect its exosphere.
- The model examines spatial and temporal variations in density and composition, considering factors like gravity, thermal desorption, radiolysis, and surface adsorption. This research helps understand how ejected materials interact with Europa's surface and environment, advancing our knowledge of its habitability.



Tidal Heating

It is a phenomenon that occurs when the gravitational forces of one celestial body cause another body to experience internal friction, generating heat in the process.

Tidal heating increases the potential for life on moons like Europa and Enceladus.

It drives volcanic and tectonic activity, shaping the surface and atmosphere of celestial bodies.

Simulation model

```
def tidal_heating(M, n, R, a, e, k):  
    """  
    Calculate tidal heating rate  
  
    Parameters:  
    M - Mass of central body (kg)  
    n - Mean motion (rad/s)  
    R - Radius of moon (m)  
    a - Semi-major axis (m)  
    e - Orbital eccentricity  
    k - Love number  
  
    Returns:  
    Tidal heating rate (W)  
    """  
    return (21/2) * M * (n)**3 * (R/a)**5 * k * e**2
```




```

class TidalHeatingSimulation:
    def __init__(self, moon_name):
        # Store moon name
        self.moon_name = moon_name

        # Moon-specific parameters
        if moon_name.lower() == 'enceladus':
            self.M_central = 5.683e26 # Mass of Saturn (kg)
            self.R_moon = 252.1e3 # Radius of Enceladus (m)
            self.a = 238.02e6 # Semi-major axis (m)
            self.e = 0.0047 # Orbital eccentricity
            self.k = 0.72 # Love number
            self.n = 5.3e-5 # Mean motion (rad/s)
            self.initial_temp = 80 # K
            self.heat_capacity = 2000 # J/kg·K (for icy material)
            self.mass = 1.08e20 # kg

        elif moon_name.lower() == 'europa':
            self.M_central = 1.898e27 # Mass of Jupiter (kg)
            self.R_moon = 1560.8e3 # Radius of Europa (m)
            self.a = 671.1e6 # Semi-major axis (m)
            self.e = 0.009 # Orbital eccentricity
            self.k = 0.37 # Love number
            self.n = 2.0548e-5 # Mean motion (rad/s)
            self.initial_temp = 110 # K
            self.heat_capacity = 2100 # J/kg·K (for icy material)
            self.mass = 4.80e22 # kg

        else:
            raise ValueError("Choose either 'Enceladus' or 'Europa'")

        self.G = 6.67430e-11 # Gravitational constant

```

```

    def tidal_heating_rate(self, t):
        """
        Calculate instantaneous tidal heating rate
        Accounts for orbital variations
        """
        # Basic tidal heating formula with time-dependent eccentricity
        time_varying_e = self.e * (1 + 0.1 * np.sin(self.n * t))
        return (21/2) * self.M_central * self.n**3 * (self.R_moon/self.a)**5 * self.k * time_varying_e

    def thermal_evolution(self, initial_temp, time_array):
        """
        Simulate thermal evolution due to tidal heating
        """
        def temp_change(T, t):
            # Calculate heat input
            heating_rate = self.tidal_heating_rate(t)

            # Temperature change rate
            dTdt = heating_rate / (self.mass * self.heat_capacity)
            return dTdt

        # Solve differential equation
        temps = odeint(temp_change, initial_temp, time_array)
        return temps.flatten()

    def simulate(self, duration=1e6, time_steps=1000):
        """
        Run full simulation
        """
        # Time array (seconds)
        time_array = np.linspace(0, duration, time_steps)

        # Simulate heating
        temperatures = self.thermal_evolution(self.initial_temp, time_array)
        heating_rates = [self.tidal_heating_rate(t) for t in time_array]

```



```

# Plotting
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 8))

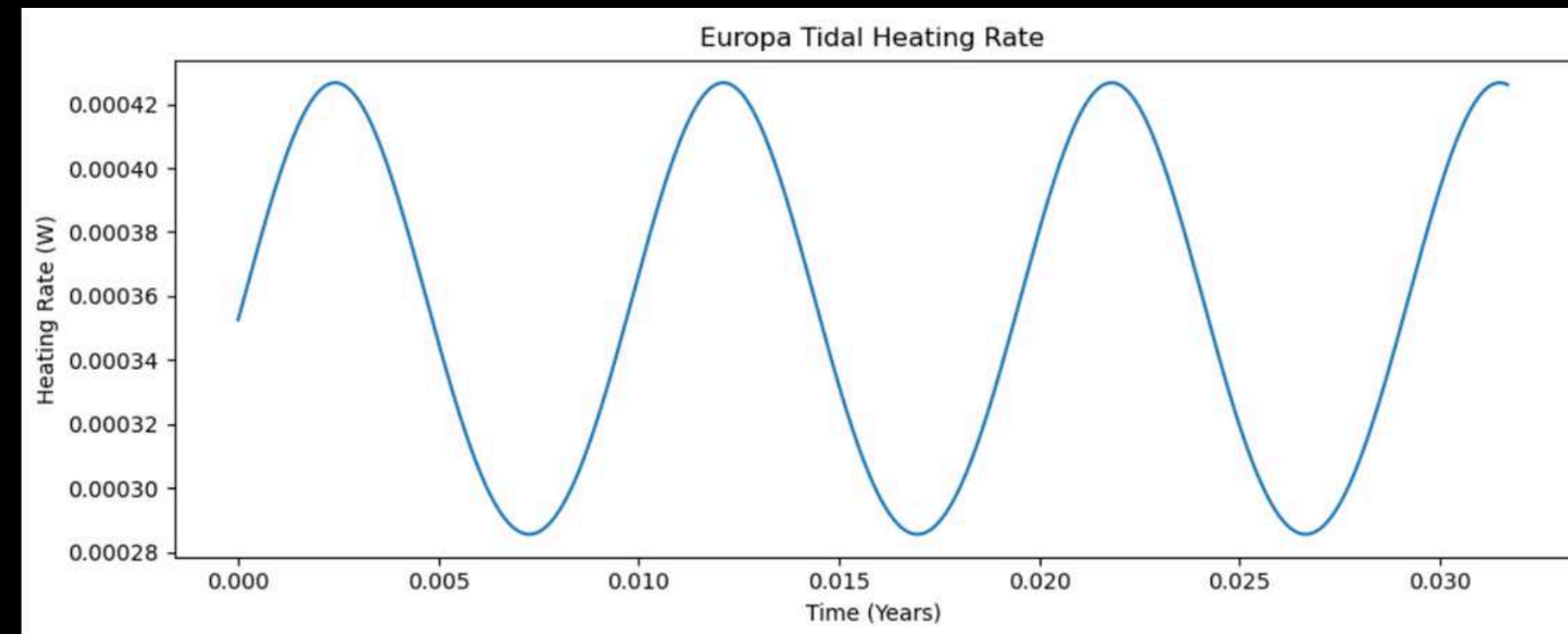
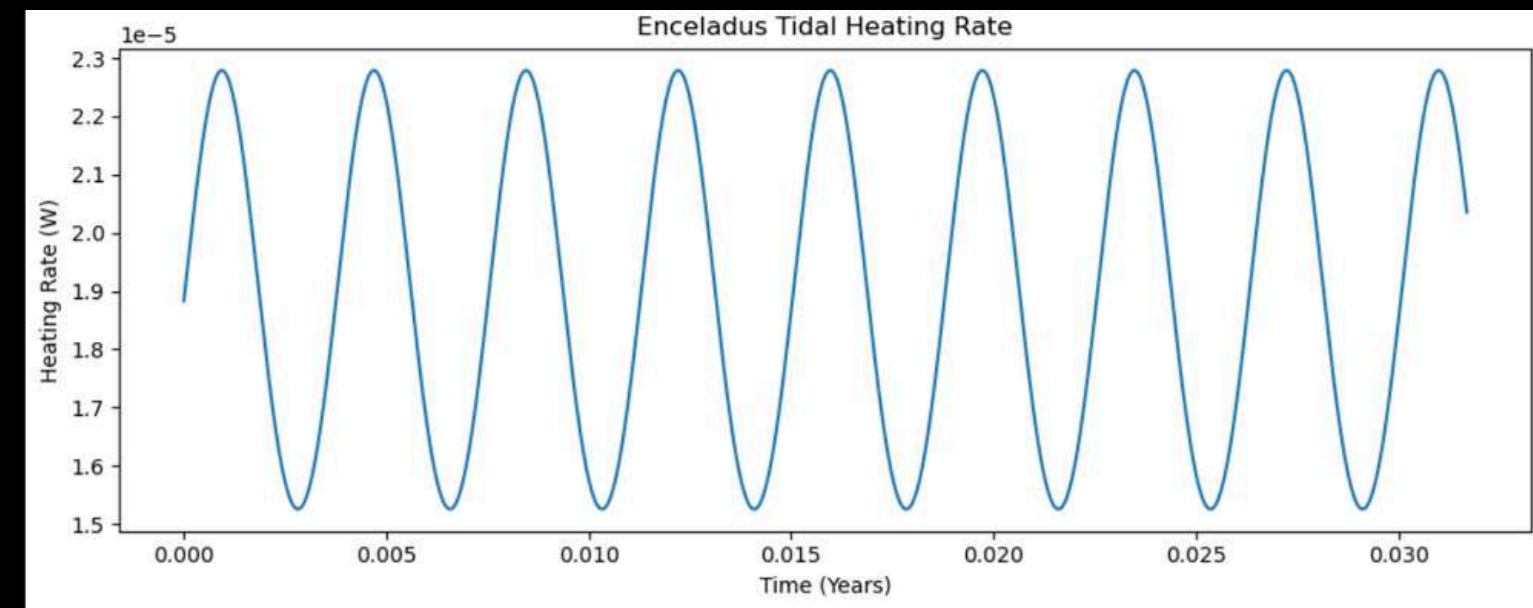
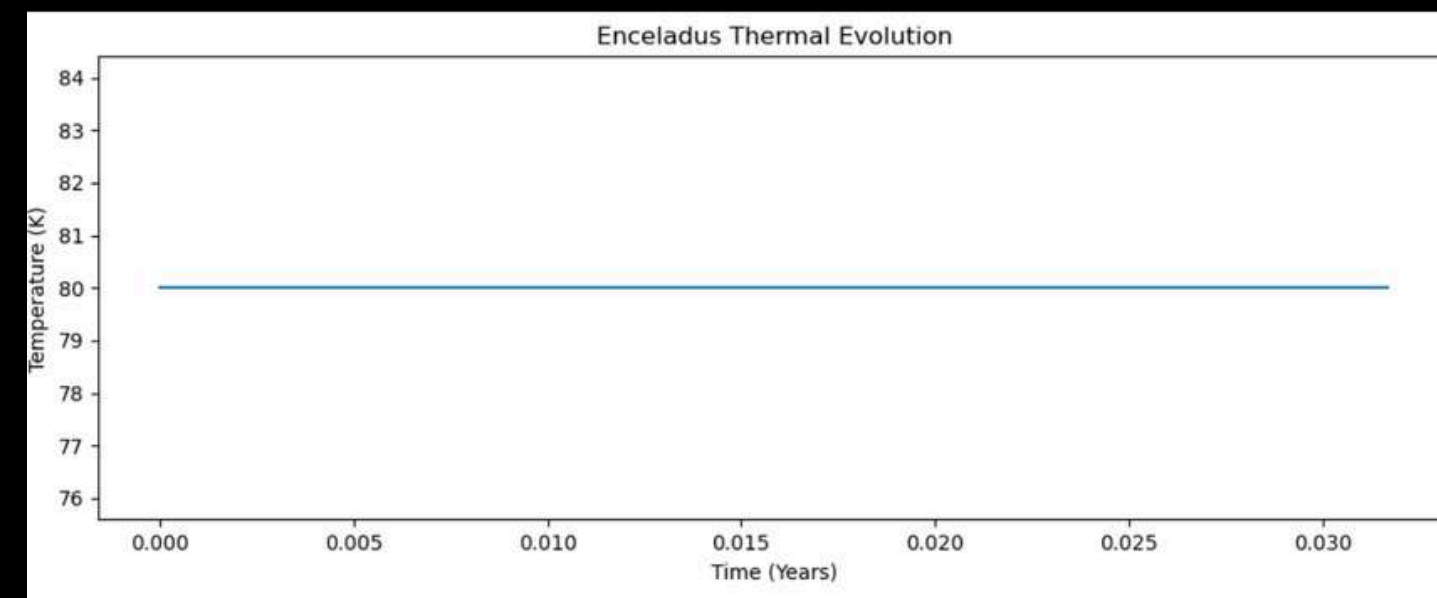
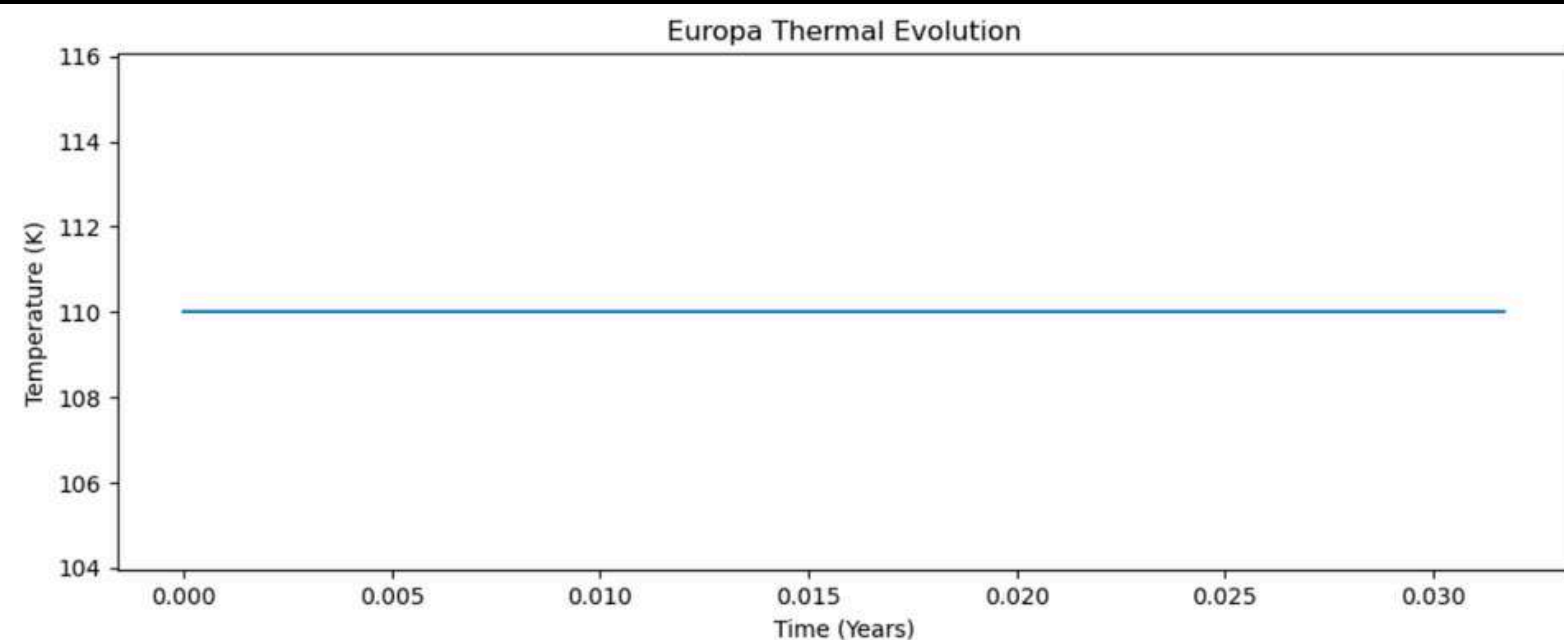
# Temperature evolution
ax1.plot(time_array / (365.25 * 24 * 3600), temperatures)
ax1.set_title(f'{self.moon_name.capitalize()} Thermal Evolution')
ax1.set_xlabel('Time (Years)')
ax1.set_ylabel('Temperature (K)')

# Heating rate evolution
ax2.plot(time_array / (365.25 * 24 * 3600), heating_rates)
ax2.set_title(f'{self.moon_name.capitalize()} Tidal Heating Rate')
ax2.set_xlabel('Time (Years)')
ax2.set_ylabel('Heating Rate (W)')

plt.tight_layout()
plt.show()

return {
    'time': time_array,
    'temperatures': temperatures,
    'heating_rates': heating_rates
}

```



SUBSURFACE TEMPERATURE

The subsurface temperature refers to the temperature beneath the outermost surface of an icy moon.

The variation of subsurface temperature is expected to take place in the following manner :

1. Near the icy surface (outer layer)

The temperature below the exposed icy layer is very low but due to insulation caused by the outer icy layers temperature is somewhat higher when compared with it.

2. Cryosphere

Temperature rise due to heat conduction from the moon's interior, as the depth within the ice shell increases. Tidal forces and radioactive decay may further elevate temperature at the base of ice.

3. Subsurface oceans

The temperature in these regions of the moons with subsurface oceans is maintained at or above the freezing point of water (273K) depending upon the pressure and salinity. Ammonia or other antifreeze compounds can lower the freezing point, maintaining liquid water at lower temperatures (e.g., ~176–273 K for Titan).

4. Rocky Core

We can find the highest subsurface temperature in rocky cores, heated by the radioactive decay and tidal friction in some cases. These temperatures can exceed 400 K near the core-ocean interface, particularly if hydrothermal activity is present (as hypothesized for Enceladus).



SUMMARY OF RESEARCH PAPERS ON SUBSURFACE TEMPERATURE AND SALINITY

Sea ice, extremophiles and life on extra-terrestrial ocean worlds

The review by Andrew Martin and Andrew McMinn explores how sea-ice microorganisms, which thrive in Earth's extreme environments, can offer insights into the potential for life on extraterrestrial ocean worlds like Europa and Enceladus. The study highlights the adaptability of extremophiles and their role in understanding habitability and the search for life on distant moons with hidden oceans.

How does salinity shape ocean circulation and ice geometry on Enceladus and other icy satellites?

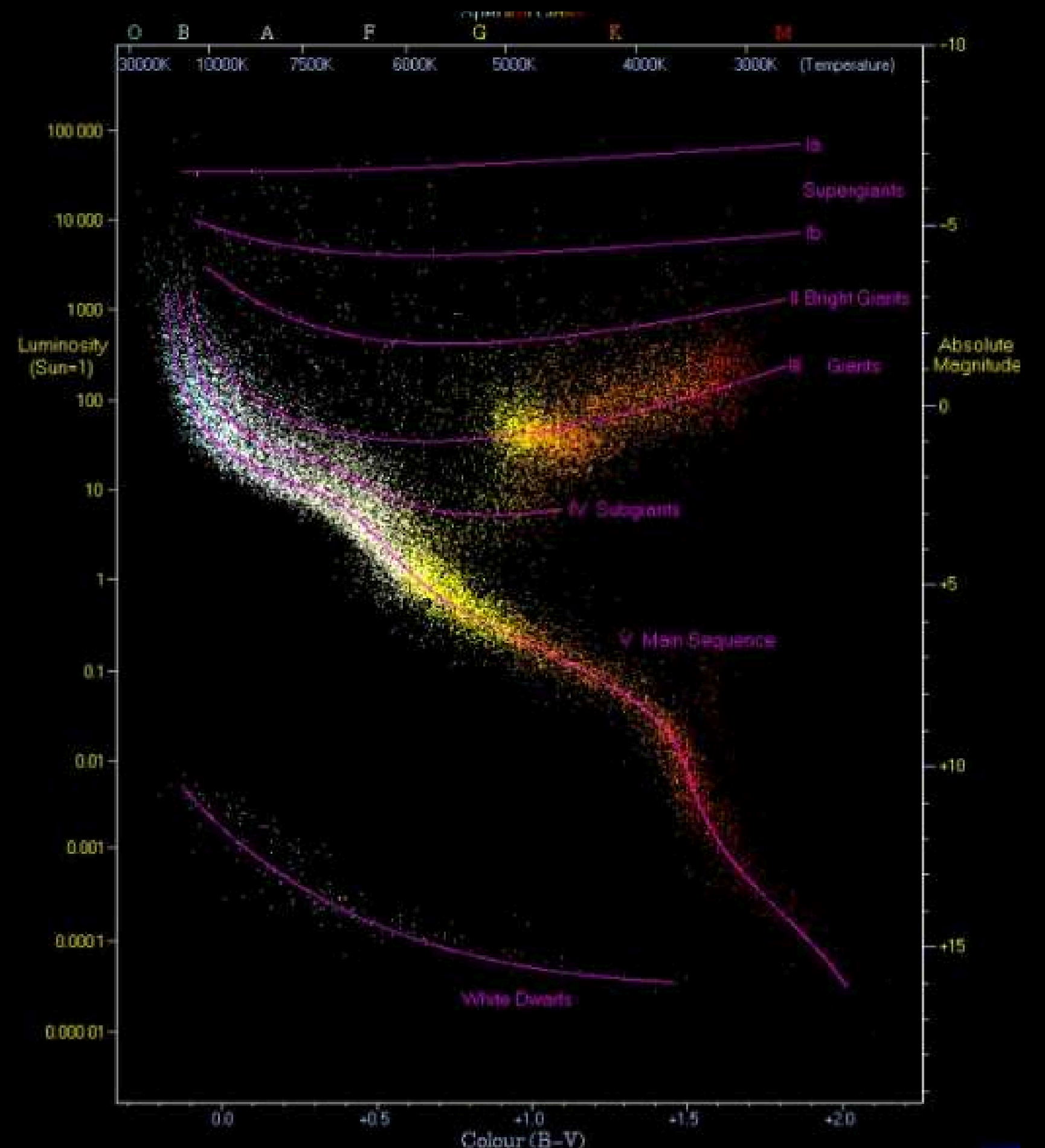
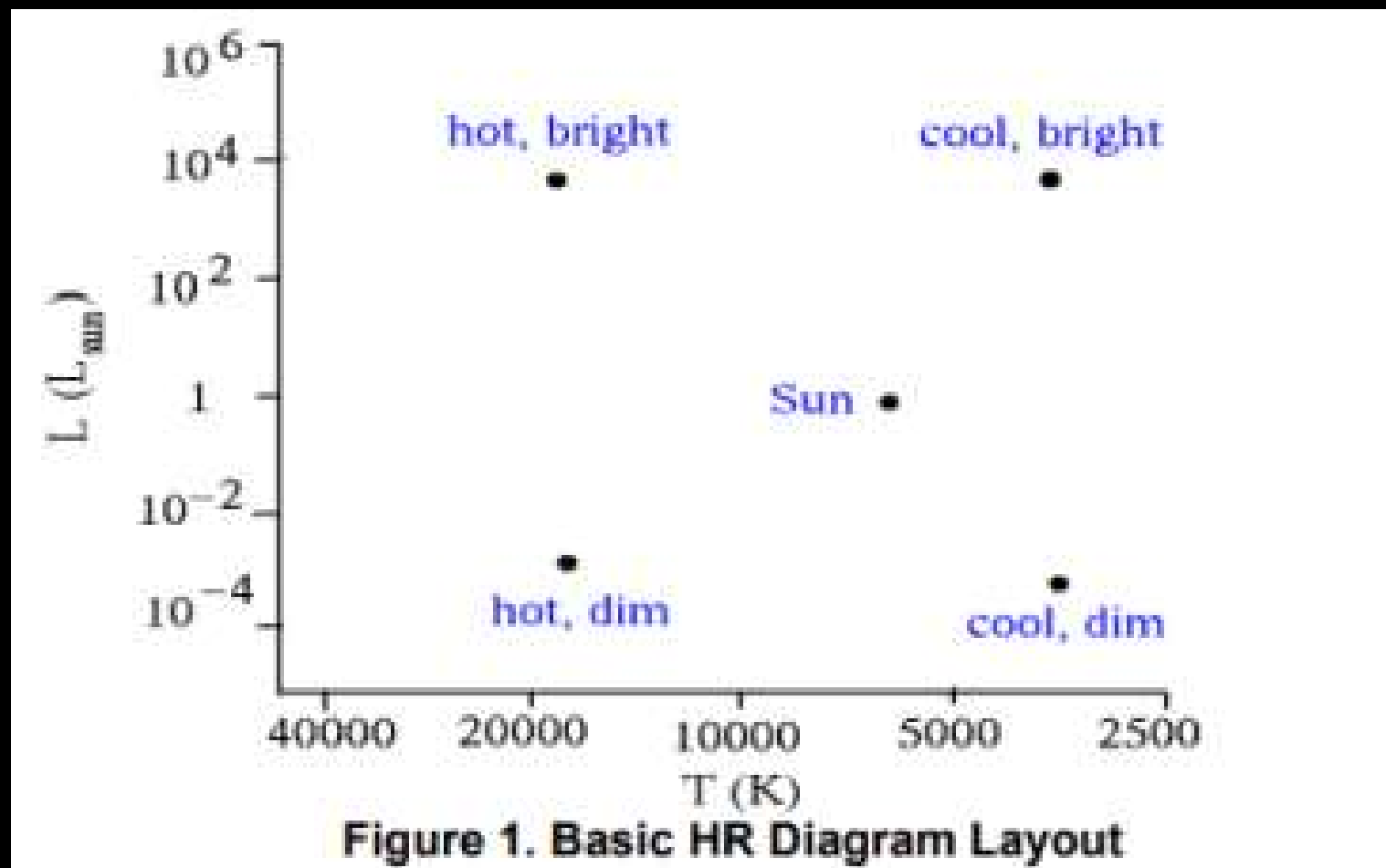
Wanying Kang's research explores how salinity affects ocean circulation and ice geometry on Enceladus, with findings showing that salinity influences heat flow and ice thickness. The study suggests that hydrothermal vents and tidal heating may keep the ocean liquid warm, and the moon's chemical environment could support life, though modeling challenges remain in understanding these dynamics.

Dynamic Europa Ocean shows transient Taylor columns and convection driven by ice melting and salinity

Yosef Ashkenazy and Eli Tziperman's study reveals that salinity drives convection and Taylor columns in Europa's subsurface ocean, influencing heat transport and ice thickness. These dynamics help explain surface features and Europa's potential for life.

THE HR DIAGRAM

Developed in the early 20th century by Ejnar Hertzsprung and Henry Norris Russell, it classifies stars based on their temperature, luminosity, and spectral characteristics.

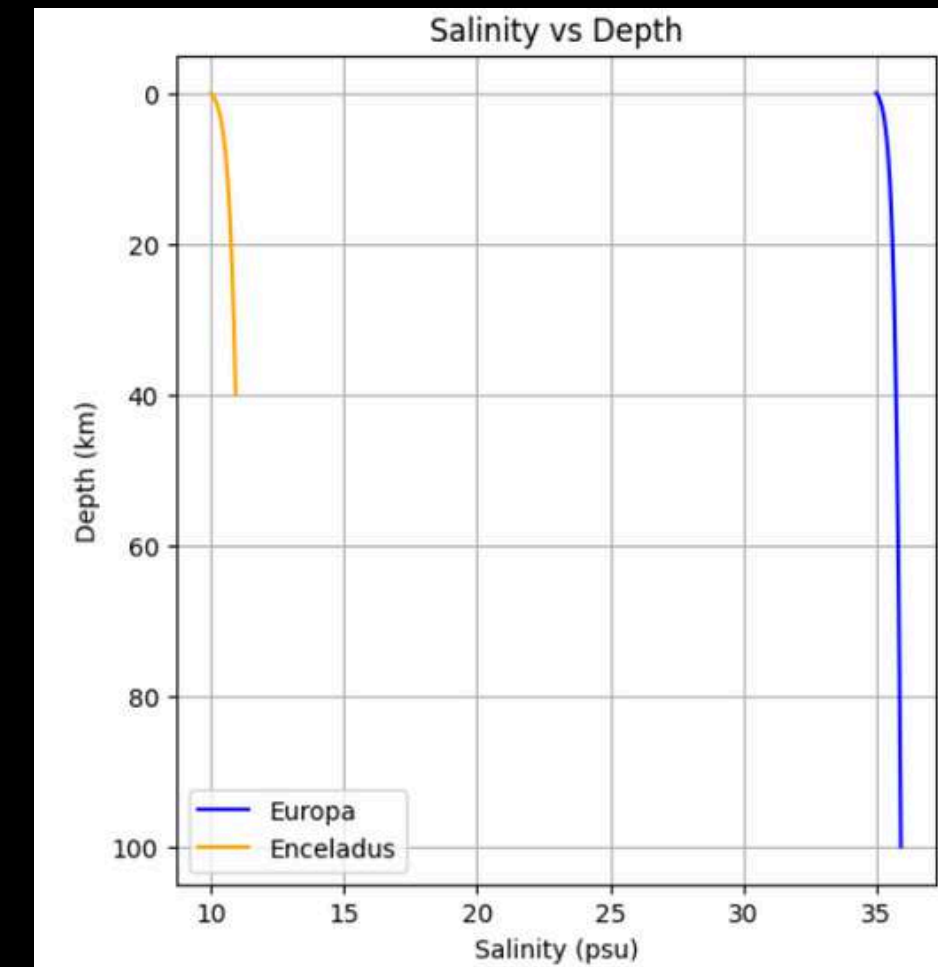
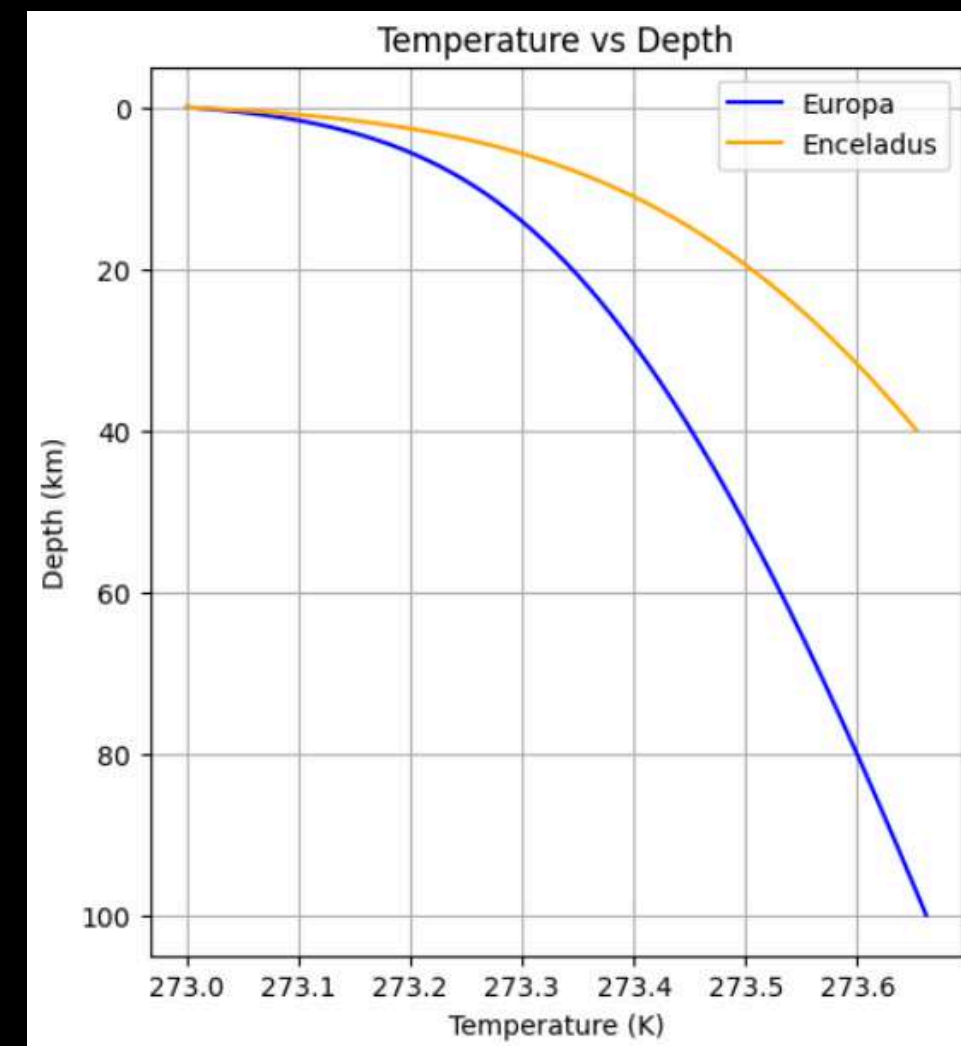



```

import numpy as np
import matplotlib.pyplot as plt
# Parameters for Europa
depth_europa = np.linspace(0, 100, 1000) # Depth in km
pressure_europa = depth_europa * 0.1 # Approximate pressure in MPa
# Adiabatic temperature gradient and MgSO4 salinity effect
temp_europa = 273 + 0.02 * pressure_europa + 0.1 * np.log(1 + depth_europa)
salinity_europa = 35 + 0.2 * np.log(1 + depth_europa)
# Parameters for Enceladus
depth_enceladus = np.linspace(0, 40, 1000) # Depth in km
pressure_enceladus = depth_enceladus * 0.08
# Hydrothermal heating effect and high pH ocean dominated by NaCl
temp_enceladus = 273 + 0.03 * pressure_enceladus + 0.15 * np.log(1 + depth_enceladus)
salinity_enceladus = 10 + 0.25 * np.log(1 + depth_enceladus)
# Plot Temperature vs Depth
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.plot(temp_europa, depth_europa, label='Europa', color='blue')
plt.plot(temp_enceladus, depth_enceladus, label='Enceladus', color='orange')
plt.gca().invert_yaxis()
plt.xlabel('Temperature (K)')
plt.ylabel('Depth (km)')
plt.title('Temperature vs Depth')
plt.legend()
plt.grid(True)

plt.subplot(1, 2, 2)
plt.plot(salinity_europa, depth_europa, label='Europa', color='blue')
plt.plot(salinity_enceladus, depth_enceladus, label='Enceladus', color='orange')
plt.gca().invert_yaxis()
plt.xlabel('Salinity (psu)')
plt.ylabel('Depth (km)')
plt.title('Salinity vs Depth')
plt.legend()
plt.grid(True)

```



EXOPLANETS

A planet that orbits a star outside our solar system (can also orbit neutron star, black hole, white dwarf or brown dwarfs).

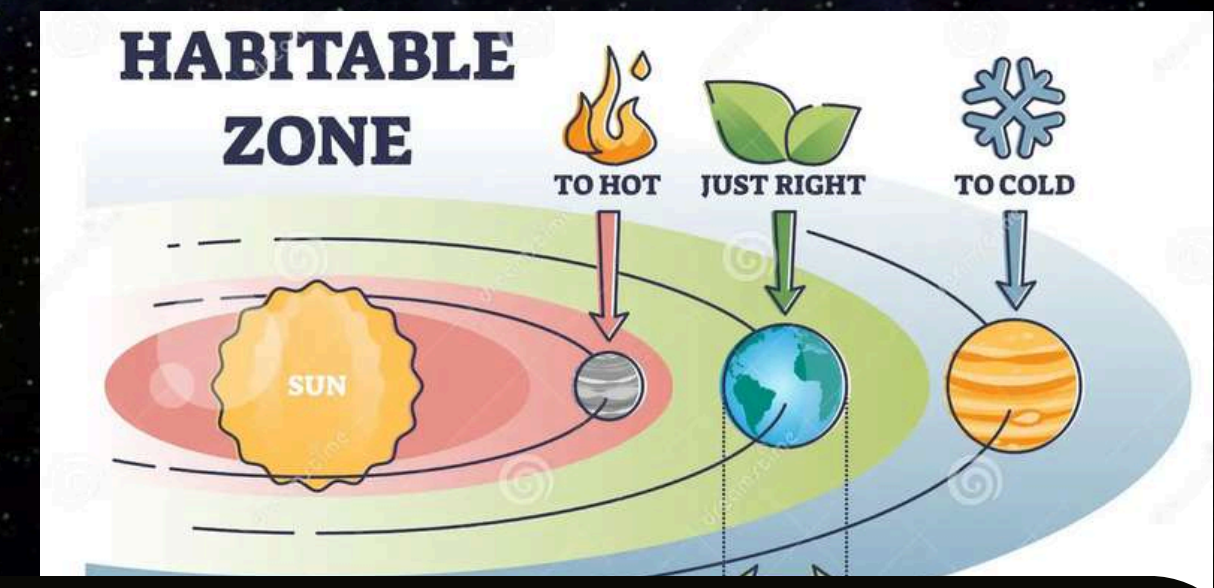
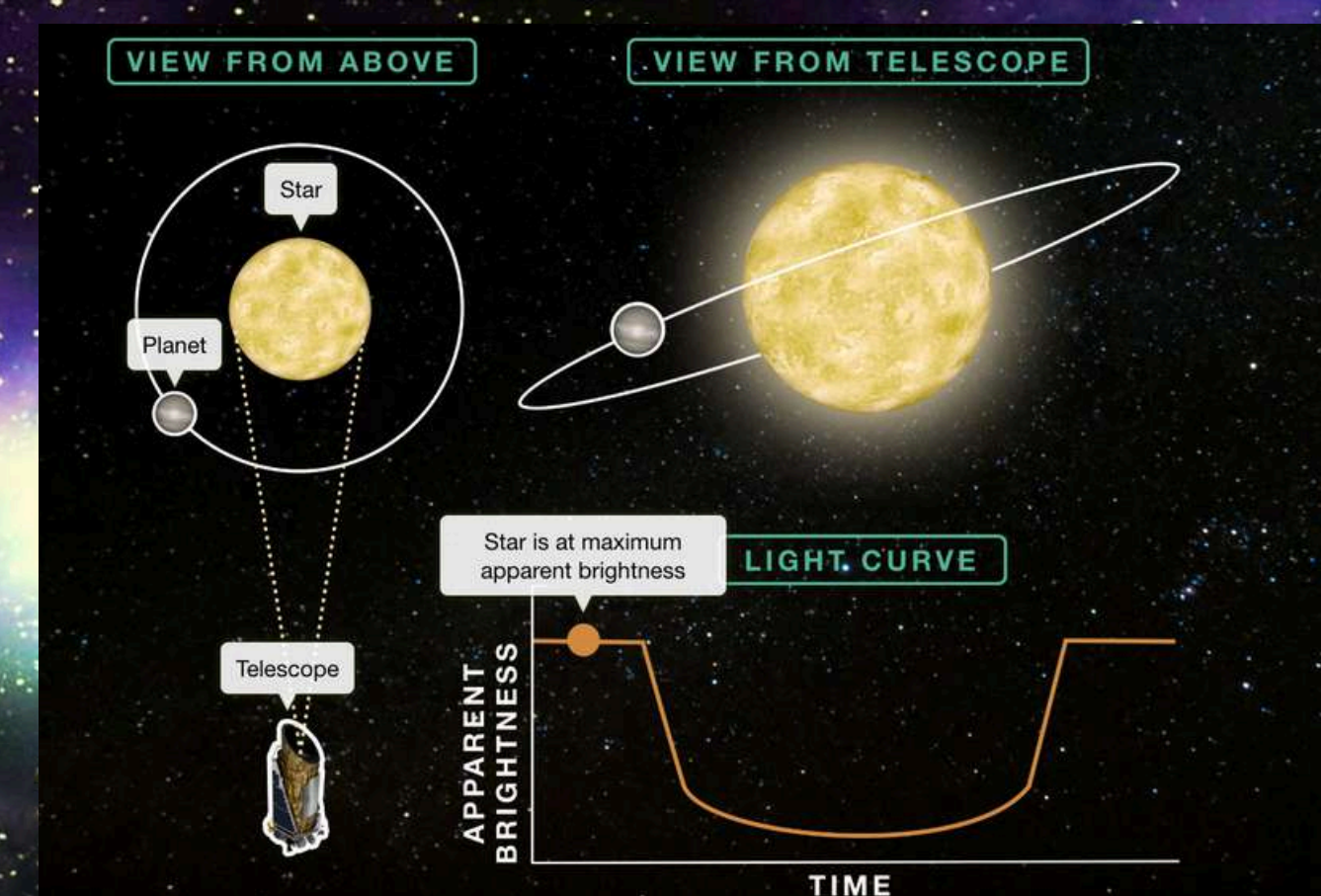
Discovery Of Exoplanets:-

- **First Discovery:** On January 9, 1992, Wolszczan and Frail discovered exoplanets orbiting the pulsar PSR 1257+12.
- **Transit Method**
 - Transit: Occurs when a planet passes in front of its star, causing a periodic dimming of the star's light.
 - Kepler's Third Law: Used to calculate planet size, orbital period, and temperature.
 - Key Insight: The dimming depth reveals the planet's size and potential habitability.

Habitability

Factors affecting habitability:

- Habitable Zone: The region around a star where liquid water could exist
- Atmosphere: A suitable atmosphere regulates temperature and protects from radiation.
- Host Star Type: Massive stars have short lifespans; red dwarfs have longer lifespans but shorter habitable zones.
- Tidal Locking: When a planet's rotation period matches its orbit, creating extreme temperature differences.
- Magnetic Field: Protects from cosmic and solar radiation, preventing atmosphere loss.
- Chemical Composition: Elements like carbon, nitrogen, oxygen, sulfur, and phosphorus are essential for life.
- Planet Size: Planets need to be large enough to retain a stable atmosphere



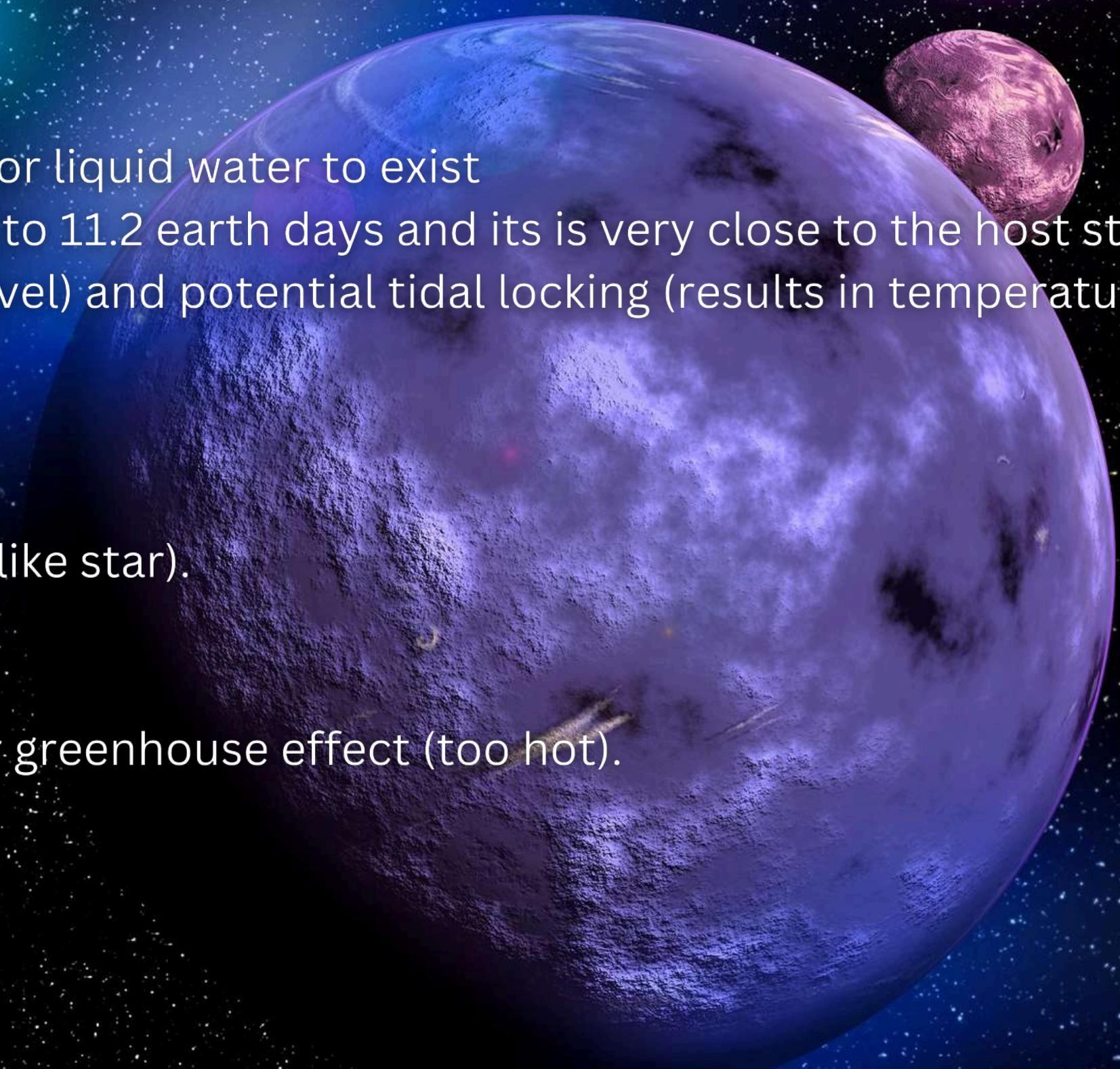
Examples of Potentially Habitable Exoplanets

- **Proxima Centauri b:**

- Star: Proxima Centauri, 4.24 light-years away.
- Size: Similar to Earth (1.17 times Earth's mass).
- Orbit: In the habitable zone which is suitable for liquid water to exist
- It's orbital time period is approximately equal to 11.2 earth days and its is very close to the host star 0.048AU
- Challenges: Stellar flares(increases radiation level) and potential tidal locking (results in temperature difference)

- **Kepler-452b (Earth's Cousin):**

- Star: Kepler-452, 1,400 light-years away (Sun-like star).
- Size: 1.6 times Earth's radius.
- Orbit: 1.04 AU (similar to Earth's orbit).
- Challenges: Larger size might cause a stronger greenhouse effect (too hot).



Habitability

Habitability refers to the ability of an environment to support life, either in its current state or with minimal modification. It encompasses the physical, chemical, and environmental conditions necessary to sustain living organisms, particularly those similar to life on Earth.

Liquid Water:

The presence of liquid water is a fundamental requirement for life as we know it. In the case of moons like Europa, Ganymede, and Enceladus, subsurface oceans beneath icy crusts are considered potential habitats for life.

Stability of the Environment:

A stable environment over time is essential for life to thrive. Europa's subsurface ocean, thought to have existed for billions of years, offering long-term conditions conducive to life.

Geological Activity:

Geological activity, like ice fracturing and internal heating, helps sustain subsurface oceans and supports potential life. Europa's active surface, observed by Voyager, hints at internal processes creating habitable conditions.

Biosignatures:

The identification of biosignatures, such as molecular markers or fossilized remains, can provide evidence of past or present life. In Europa's case, scientists are particularly interested in how materials from the ocean might rise to the surface, potentially revealing biological activity.

Temperature:

Appropriate temperatures are vital for liquid water and life. Tidal and radiogenic heating on icy helps moons like Europa to sustain liquid water and create stable, potentially habitable environments.

Biogenic Elements:

The availability of essential biological components such as carbon, nitrogen, phosphorus, and sulfur is crucial for supporting life. These elements are believed to be present in environments like Europa's ocean, potentially supporting life forms.

Radiation:

The level of radiation on the surface of a celestial body impacts the habitability of its subsurface environment. While Europa's surface is exposed to radiation, its ice shell may protect any subsurface life, and ongoing research aims to understand the radiation levels better.

Energy Sources:

A reliable energy source is vital to sustain biological processes. Europa's subsurface ocean, for instance, could provide energy through thermal and chemical interactions between the ocean and the icy crust, as well as hydrothermal vents that offer energy to support metabolism.

ESI

- The Earth Similarity Index (ESI) is a multiparameter metric that quantifies how similar a planet, moon, or celestial body is to Earth, expressed as a value between 0 (no similarity) and 1 (identical to Earth).
-
- It serves as a fuzzy comparison rather than a direct measure of habitability, using a distance metric based on selected planetary properties and their corresponding Earth values.
-
- ESI calculations improve with more properties, but determining these characteristics can be challenging due to limitations in detection methods. The ESI applies to planets and small celestial bodies, like Titan.



[0.91]



[0.89]

$$ESI = \prod_{i=1}^n \left(1 - \left| \frac{x_i - x_{i0}}{x_i + x_{i0}} \right| \right)^{\frac{w_i}{n}},$$



[0.87]



[0.86]

ESI Prediction Model

```
features = [  
    'Radius_km', 'Surface_Temp_K', 'Albedo', 'Surface_Gravity_m_s2',  
    'Escape_Velocity_km_s', 'Radiation_mSv_day', 'Tidal_Heating_GW'  
]
```

Firstly some important features related to habitability are separated out from the dataset.

```
# Splitting data into features (X) and target (y)  
X_known = known_esi_data[features]  
y_known = known_esi_data['ESI_Global']  
  
# Filling missing feature values with the column mean (if any)  
X_known = X_known.fillna(X_known.mean())  
X_unknown = unknown_esi_data[features].fillna(X_known.mean())
```

Next, the dataset is broken into features and targets ie. ESI Value. Empty fields are then filled using the mean of data of that field across the dataset of known ESI Data

```
# Use the imputed column for training the Random Forest model  
X_full = data[features].fillna(X_known.mean())  
y_full = data['Imputed_ESI_Global']  
  
# Train-test split for evaluation  
X_train, X_test, y_train, y_test = train_test_split(X_full, y_full, test_size=0.3, random_state=42)  
  
# Train the Random Forest Regressor  
random_forest = RandomForestRegressor(random_state=42, n_estimators=100)  
random_forest.fit(X_train, y_train)  
  
# Predict on the test set and evaluate  
y_pred = random_forest.predict(X_test)  
mse = mean_squared_error(y_test, y_pred)  
r2 = r2_score(y_test, y_pred)  
  
# Predict missing ESI_Global values using the trained Random Forest model  
rf_predictions = random_forest.predict(X_full[data['ESI_Global'].isnull()])  
  
mse, r2, rf_predictions
```

The Data is broken into train and test data. the data is used to train a random forest Regressor. after which it is asked to make prediction on the test data which is checked against the existing result. MSE and R2 Score are then measured to check the accuracy

Active Missions on Jupiter



Juno (NASA):

- Launched on August 5, 2011;
- arrived at Jupiter on July 4, 2016.
- Extended mission runs until 2025.
- Key discoveries include chaotic storms, Earth-sized polar cyclones, and a fuzzy core
- It highlighted unique ammonia-ice lightning and weather
- contrasts with Earth.



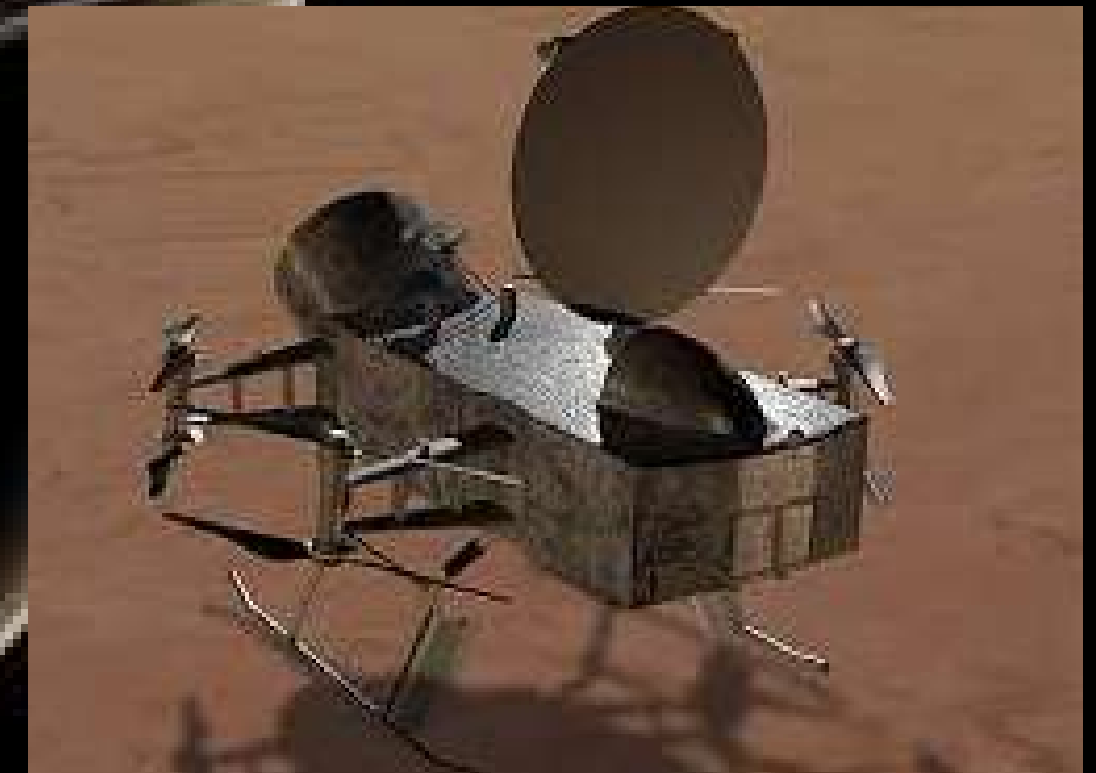
Juice (ESA):

- Launched in April 2023.
- Focused on Jupiter's icy moons—Ganymede, Europa, and Callisto.
- Studies include surface compositions, subsurface oceans, habitability, and magnetic interactions.

Missions on Saturn

Dragonfly (NASA):

- Scheduled to launch in 2028, targeting Saturn's moon Titan.
- The drone-like craft will explore Titan's methane-rich atmosphere, organic molecules, and surface features like dunes and craters.
- It aims to study Titan's chemistry and potential for life.



Europa Clipper Mission

Goal-The primary motive of the mission was explore Europa and study its subsurface ocean .

It was launched on October 14, 2024 and is foresighted to reach the orbit by April 2030.

The spacecraft will orbit Jupiter and conduct 49 close flybys of Europa, gathering data on its surface, ice shell, and oceans.

Cassini Ion and Neutral Mass Spectrometer (INMS) Data

- Plume Composition: Dominated by water vapor (H_2O) (~91%).
- Contains significant amounts of carbon dioxide (CO_2) (~3.2%), methane (CH_4) (~1.6%), and an unidentified species with a mass-to-charge ratio of 28 (likely N_2 or CO).

Plume Dynamics:

The structure is asymmetrical, with spatial and temporal variations

Trace amounts of ammonia (NH_3), acetylene (C_2H_2), and propane (C_3H_8) were also found

Implications for Habitability:

The subsurface environment beneath the plume likely supports liquid water

The Artemis program

The Artemis program was led by NASA and initiated in 2017. It is aimed at returning humans to the Moon and establishing a sustainable presence there.

Missions:

- Artemis I (2022):
An uncrewed mission testing the Orion spacecraft and SLS rocket.
- Artemis II (planned for 2026):
The first crewed mission to fly around the Moon.
- Artemis III (planned for 2027):
The program's first crewed lunar landing near the Moon's south pole.



Key Features:

- Orion Spacecraft
Designed for deep-space exploration, it provides crew transport to lunar orbit and back to Earth.
- Space Launch System (SLS)
SLS is NASA's most powerful rocket, built to propel Orion and heavy payloads into space
- Lunar Gateway
- Stellar Navigation (star navigation)
- Incorporation with advanced AI and voice technology



FUTURE EXPLORATION

A full-page background image showing an astronaut in a spacesuit walking on the lunar surface. The astronaut is in the foreground, slightly out of focus, with their legs and boots visible. In the background, the Earth is visible in the black sky, and a lunar lander or base structure is partially visible on the horizon.

Artemis Program: The Artemis program aims to return humans to the Moon by the mid-2020s, including the first woman and the next man. The goal is to establish a sustainable presence on the Moon and use it as a stepping stone for future Mars missions

International Collaboration: Future Moon exploration efforts involve significant international collaboration, with various countries planning missions that include lunar bases, research stations, and resource extraction initiatives.

Conclusion

Through this project we learnt about the geophysical characteristics of the icy moons Europa and Enceladus, two of the potentially most habitable celestial bodies in the solar system. We learnt how their geological phenomena affect the temperature and other parameters on them, thereby affecting the potential for habitability.

We also proceeded analytically and used computation to relate temperature, salinity and depth; understand variation of thermal properties with time, and also used a machine learning model to calculate the ESI of icy moons. We then explored the ongoing and upcoming space missions to these bodies.

Overall we gained a good insight into planetary geology and also touched up a bit on astrobiology, exploring our moon as well as the icy moons of the solar system.

Thank You!